SKIN Cancer: Detection and Color Bias

Arvind Mahalingam
Ian Wang
Kenny Gao

TABLE OF CONTENTS

01 Dataset

02 Model

03 Methodology

04 Stats. Analysis

05 Improvement

Dataset

SLICE-3D Dataset

2D image crops from 3D Total Body Scans

- ~400,000 training images of varying lesion size and skin tones

Problem Statement

- Implement a model that predicts skin lesions and extrapolate model performance relative to different categories
  - see if the model also inherits bias
  - see if we can decrease its bias (if there is any)

Data Augmentation

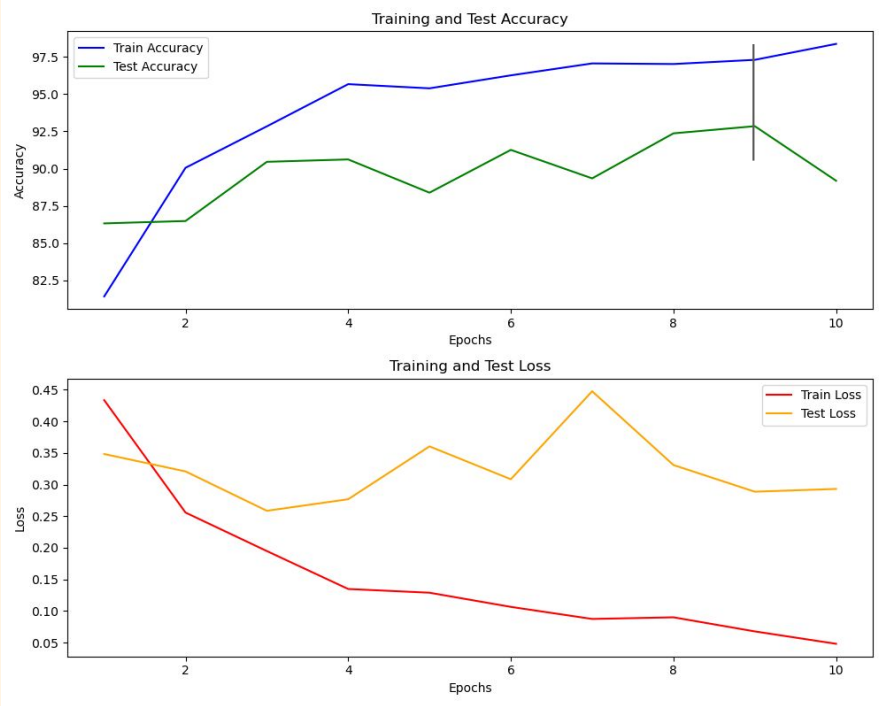
- Before: Unbalanced Dataset
  - 393 / 401059 are labeled malignant
- Augmentation
  - Horizontal flip
  - Vertical flip
  - 180° Rotation
- After: Balanced Dataset
  - 1572 malignant V.S. 1572 benign

Model

Efficient Net

- We choose pre-trained Efficient Net for its size/performance
  - smaller is better
- Hyperparameters and other implementation details
  - swapped out the final layer for a binary classifier
  - 10 epochs, 0.001 lr, AdamW
  - final model based on epoch with best test acc.
- train model, run predictions, run stats

Model Performance



- 92.5% Test Acc
- overfitting
- best performance by unfreezing last 2 layers

Methodology

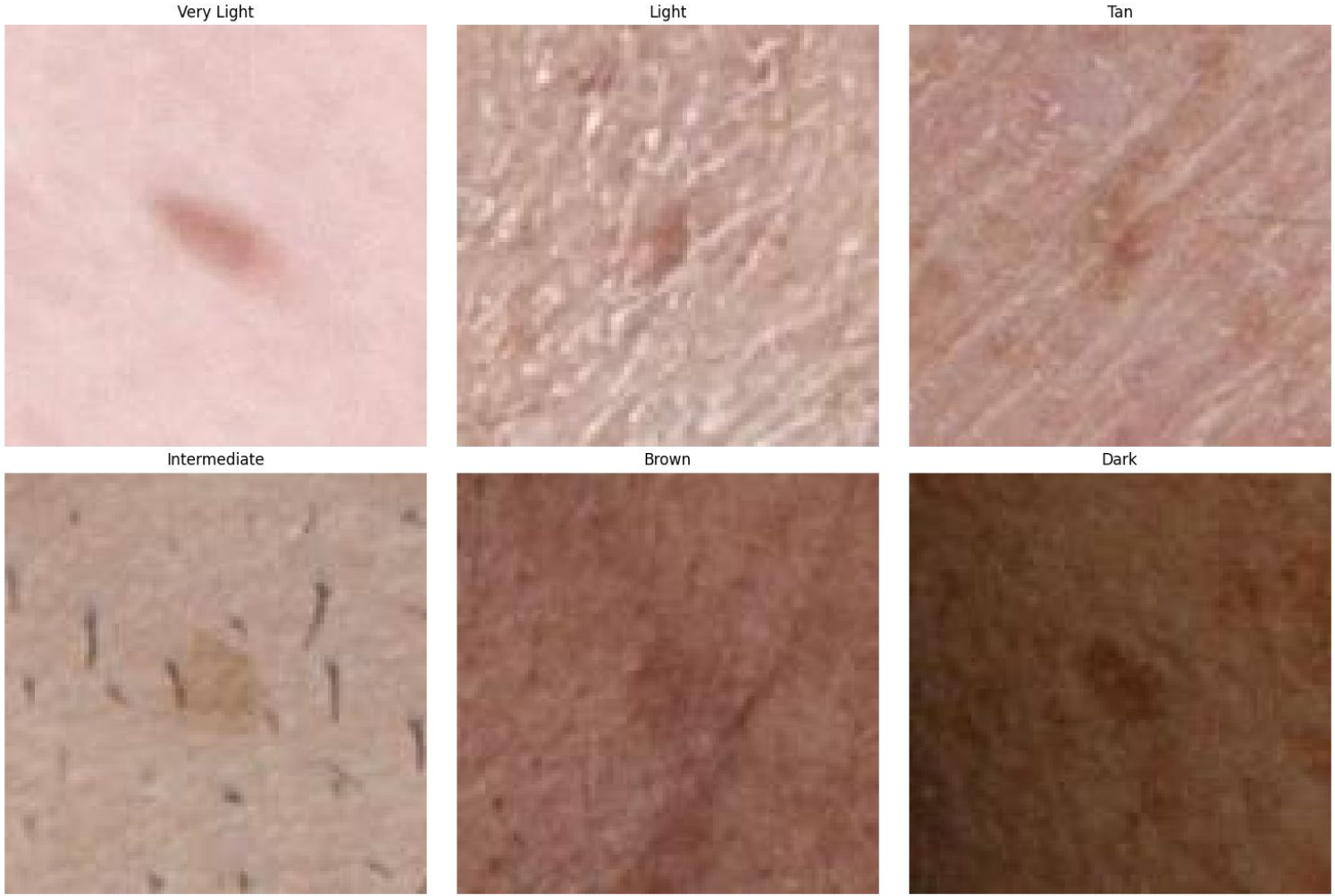
Statistical Methodology

- **Henry Ford Health.** (2021, November 19). *Individual Typology Angle and Fitzpatrick Skin Phototypes are not equivalent in photodermatology.* Henry Ford Health Scholarly Commons.
- Formula
  - ITA° = [arctan(L*-50)/b*)] *180/p
  - L is lightness outside the legion
  - B is the Blue-Yellow axis outside

    the legion

Table 1B. Individual typology angle (ITA) skin classification proposed by Chardon *et al* (12).

Reference ITA skin classification (12)	ITA° range
Very light	ITA° > 55
Light	41 < ITA° < 55
Intermediate	28 < ITA° < 41
Tan	10 < ITA° < 28
Brown	−30 < ITA° < 10
Dark	ITA° < −30


Very Light
Light
Tan
Intermediate
Brown
Dark

Statistical Methodology (cont.)

- We initially used an ANOVA test then we settled on chi-squared
- We also tried the Kruskal-Wallis test
- We also tested on different categories such as gender and age
  - We used chi squared tests for both as well
- We tested on malignancy rate for the original data set, and we tested on error rate on the learner data set among different categories

Statistical Analysis

Research Question

- Our research question was whether or not our machine learning model mirrored real world biases concerning diagnosing dark skin after training
- Our null hypothesis was that all of the skin color categories would have an equivalent error rate
- Our alternate hypothesis was that all of the skin color categories did not have an equivalent error rate
    - With the assumption that lighter skin colors would have a lower error rate, corroborating with research

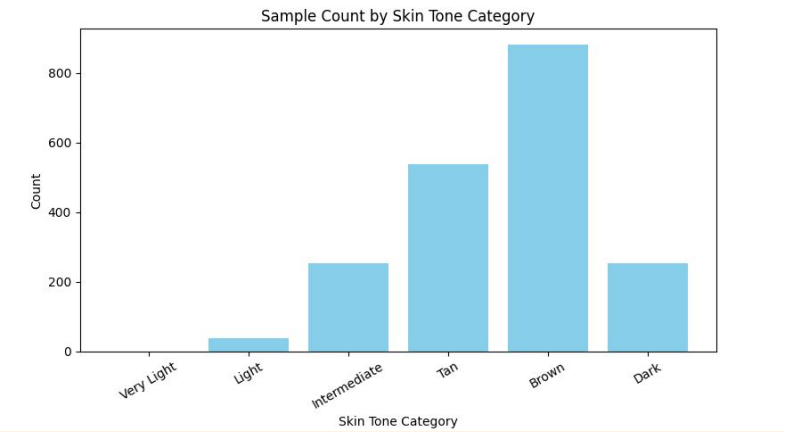
Original Data Set

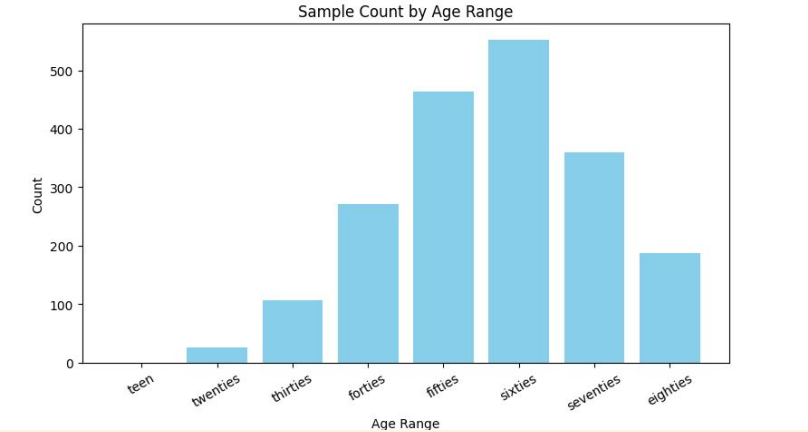
Basic Analysis

```
Malignant lesion percentage by skin tone:

skin_tone
Very Light        NaN
Light            7.69
Tan             21.79
Intermediate    20.16
Brown           19.25
Dark            20.55
Name: malignant_lesion_percentage, dtype: float64

Overall Malignant lesion percentage:
20.0%
```





Testing on Skin Tones

Chi-Squared test:

```
Chi-Square Statistic: 5.1256
P-value: 0.2746
Degrees of Freedom: 4
Expected Frequencies:
[[ 31.2   7.8]
 [429.6 107.4]
 [202.4  50.6]
 [706.4 176.6]
 [202.4  50.6]]

Groups skipped due to insufficient data (< 2 non-null target values):
- Very Light
```

Kruskal-Wallis test:

```
Kruskal-Wallis H-statistic: 5.1230
P-value: 0.2749

Groups skipped due to insufficient data (< 2 non-null target values):
- Very Light
```

Tests on different categories

Chi Squared - Age ranges:

```
F-statistic: 7.2777
P-value: 0.0000001

Groups skipped due to insufficient data (< 2 non-null target values):
- teen
```

Chi Squared - Gender:

```
Chi-squared Statistic: 3.5077
P-value: 0.0611
```

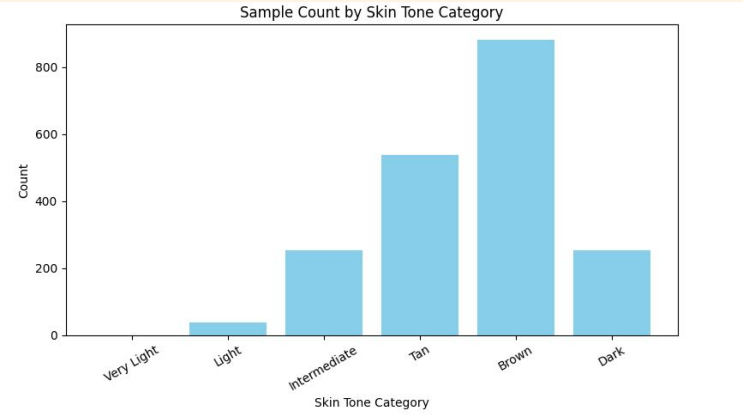
Learner Data Set

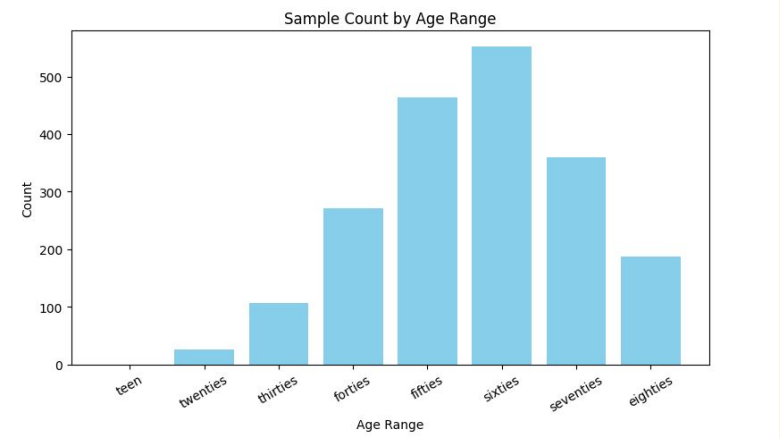
Basic Analysis

```
Error percentage by skin tone:

skin_tone
Brown           14.95
Dark            16.21
Intermediate    18.58
Light            2.56
Tan             19.74
Name: error_percentage, dtype: float64

Overall error percentage:
16.64%
```





Testing on Skin Tones

Chi Squared test:

```
Chi-Square Statistic: 11.8276
P-value: 0.0187
Degrees of Freedom: 4
Expected Frequencies:
[[ 32.50992366   6.49007634]
 [447.63664122  89.36335878]
 [210.89770992  42.10229008]
 [736.05801527 146.94198473]
 [210.89770992  42.10229008]]

Groups skipped due to insufficient data (< 2 non-null target values):
- Very Light
```

Kruskal-Wallis test:

```
Kruskal-Wallis H-statistic: 4.0339
P-value: 0.4014

Groups skipped due to insufficient data (< 2 non-null target values):
- Very Light
```

Tests on different categories

Chi squared - Age ranges:

```
Chi-Square Statistic: 41.9807
P-value: 0.00000019
Degrees of Freedom: 6
Expected Frequencies:
[[ 20.83757637   4.16242363]
 [ 88.35132383  17.64867617]
 [225.8793279   45.1206721 ]
 [386.74541752  77.25458248]
 [460.09368635  91.90631365]
 [299.22759674  59.77240326]
 [155.86507128  31.13492872]]

Groups skipped due to insufficient data (< 2 non-null target values):
- teen
```

Chi squared - Gender:

```
Chi-squared Statistic: 0.6453
P-value: 0.4218
```

Conclusion!

- Because the p value on the chi squared test was below the threshold value of 0.05, we reject the null hypothesis, meaning there is evidence that our learned model inherited racial biases
- The Kruskal-Wallis test says otherwise, but that is most likely because the variables are categorical and not continuous
- This shows the importance of alleviating underlying biases when it comes to image based machine learning, and how left unaddressed it is significant

Improvement

Future Improvement

- More even skin tone distribution
  - Weighted random sampling
- Or use a weighted loss function
- Lazy loading from HDF5 file
- Explore a different architecture, such as EfficientNetV2, ConvNeXt

Sources

- Henry Ford Health. (2021, November 19). *Individual Typology Angle and Fitzpatrick Skin Phototypes are Not Equivalent in Photodermatology*. Henry Ford Health Scholarly Commons. https://scholarlycommons.henryford.com/dermatology_articles/528\
- https://www.jaad.org/article/S0190-9622(21)00653-8/fulltext
- https://www.sap.com/resources/what-is-ai-bias#:~:text=Bias%20in%20AI%20models%20typically,used%20to%20train%20the%20AI.
- International Skin Imaging Collaboration. (2024). *SLICE-3D 2024 Challenge Dataset*. International Skin Imaging Collaboration. https://doi.org/10.34970/2024-slice-3d

- Licensed under CC BY-NC 4.0.

- Images from: Hospital Clinic de Barcelona, Memorial Sloan Kettering Cancer Center, Hospital of Basel, FNQH Cairns, The University of Queensland, Melanoma Institute Australia, Monash University and Alfred Health, University of Athens Medical School, and Medical University of Vienna.

Thank you!

Statistical Methodology

- We used ITA skin color classification rather than the Fitzpatrick classification as the Fitzpatrick classification is subject, and we could assemble the ITA skin color classification using information from the given data set
- We got our formula and our boundaries for classification from the research paper “Individual Typology Angle and Fitzpatrick Skin Phototypes are Not Equivalent in Photodermatology” by Muhammed Osto et al.
- Formula
  - ITA° = [arctan(L*-50)/b*)] *180/p
  - L is lightness outside the legion
  - B is the Blue-Yellow axis outside the legion

Table 1B. Individual typology angle (ITA) skin classification proposed by Chardon *et al* (12).

Reference ITA skin classification (12)	ITA° range
Very light	ITA° > 55
Light	41 < ITA° < 55
Intermediate	28 < ITA° < 41
Tan	10 < ITA° < 28
Brown	−30 < ITA° < 10
Dark	ITA° < −30

Statistical Methodology (cont.)

- Initially ANOVA test, but ended up with chi-square test
  - the data is binary and not normally distributed
- Also tried the Kruskal-Wallis test,
  - No normal distribution distribution assumption, requires continuous,
  - to see if the results were different
- Tested on different categories such as gender and age,
  - to see if there was there was any significance there.
  - Chi squared tests for both as well
- We tested on whether or not the lesion was malignant or benign for the original data set, and we tested on error rate on the learner data set